

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit: Department of Electronics and Electrical Communication Engineering**

2. **Subject Name: Digital Signal Processing- II L-T-P: 3-1-0 Credits: 4**

3. **Pre-requisites: Digital Signal Processing- I**

4. **Syllabus and reference books:**

Syllabus:

Multirate Signal Processing, Decimation and interpolation, analysis and synthesis filter banks; QMF, M-Channel filter banks; paraunitary filter banks and perfect reconstruction; Wavelets and Multiresolution Signal Analysis; multiresolution decomposition of signal space, analysis-synthesis filter banks, orthonormal wavelet filter banks, Haar wavelets, Daubechies and spline wavelets; time frequency decomposition, short term Fourier transform and Wigner-Ville distribution. Sparse Signal Processing, compressive sensing, l_1 -norm regularization, basis pursuit, restricted isometry property, greedy algorithms. Graph Signal Processing, motivation from big data and large scale sensor network applications; adjacency and Laplacian matrix, graph Fourier transform, convolution on graphs, filter banks on graphs.

Reference Books:

1. Multirate Systems and Filter Banks: P. P. Vaidyanathan, Pearson Publication, 2006
2. Wavelets and Filter Banks: Gilbert Strang and T. Nguyen, Wellesley Cambridge Press, 1996
3. Foundations of Signal Processing: Martin Vitterli, Jelena Kovacevic and Vivek Goyal, Cambridge University Press, 2014
4. Wavelets and subband coding: Martin Vitterli and Jelena Kovacevic, Prentice Hall, 1995
5. A wavelet Tour of signal Processing- The sparse way: Stephane Mallat, Elsevier, 2014
6. A Mathematical Introduction to Compressive Sensing: Simon Foucart and Holger Rauhut, Springer, 2013
7. Introduction to Graph Signal Processing: Antonio Ortega, Cambridge University Press, 2022.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Multirate Signal Processing : Decimation and interpolation of sequences, Noble identity and efficient decimator and interpolator;	4
2.	Filter banks: Analysis and synthesis filter banks; uniform DFT and DCT filter banks; QMF, M-Channel filter banks; paraunitary filter banks and perfect reconstruction; examples from sub-band coder, transmultiplexers.	6
3.	Wavelets and Multiresolution Signal Analysis: Basic scaling function, mother wavelet, multiresolution decomposition of signal space, analysis-synthesis filter banks induced by multiresolution analysis.	6
4.	Orthonormal wavelet filter banks, construction of special wavelets like Haar wavelets, Daubechies and spline wavelets;	6
5.	Time frequency Analysis: Time analysis, frequency analysis, and scale analysis; time-bandwidth relations; instantaneous frequency; densities and local quantities; the short time Fourier Transform; the Wigner-Ville representation; general joint representations; higher-order time-frequency distributions.	6
6.	Sparse Signal Processing: Examples of sparse signals and systems, compressive sensing of sparse signals, l_1 norm regularization, restricted isometry property of sensing matrices, compressive sampling.	6
7.	Sparse Recovery Algorithms: basis pursuit, greedy algorithms, orthogonal matching pursuit, compressive sampling matching pursuit, subspace pursuit, iterative hard thresholding and hard thresholding pursuit, applications.	6
8.	Graph Signal Processing : Motivation from big data and large scale sensor network applications; graph structure and forms, adjacency and Laplacian matrix of a graph, spectral graph theory, concept of graph Fourier transform, graph shift operator, convolution on graphs and graph filters, subsampling and filter banks on graphs, graph random processes.	8
Total number of hours		48